Our extended reality truly begins in 2024

Leave AR and VR behind in 2023, start saying 'extended reality' or XR for all things holographic, semi-immersive or fully immersive visual experiences from 2024 and beyond. Why do I say this, when things like the Meta Quest headset, Snapchat's Spectacles and Microsoft HoloLens (among others) have existed for a few years now? Allow me to explain.

Just for comparison, think about the smartphone or foldable phones wave. One can say that smarter feature phones did exist up until 2007, just as the industry was preparing for an impending Android phone launch, but it wasn't until Apple released the original iPhone did the current smartphone race truly kickoff. Similarly, the first commercially available foldable screen smartphone debuted in 2018, but it took until 2023 for every major smartphone maker (except Apple) to have a foldable variant out – all of them officially selling in India. Today, everyone acknowledges the iPhone as the foundation of the current smartphone industry, and 2023 will forever be remembered as a special year in the history of foldable phones. For the XR industry, it will be 2024, for no small part played by the Apple Vision Pro.

The Cupertino giant perhaps considers foldables as a flash in the pan compared to the limitless possibilities of XR-based experiences. There can be no arguments against the fact that Apple's brand recognition and user base are immense. If the Vision Pro proves successful and user-friendly, it could drive mass adoption of XR technology like nothing else. Unlike traditional VR headsets, Vision Pro appears to prioritise spatial computing with enhanced phygital experiences. This could potentially spur development of spatial computing. On the technical front, Apple's focus on high-end, integrated hardware and software is likely to inspire a lot of industry firsts as far as XR experiences go. From an ecosystem, Apple's roadmap of releasing tools and APIs for VR/AR development could benefit the entire industry, creating a more standardised platform.

If Vision Pro is destined to be the iPhone moment of the XR industry, let's just say the competition like Google, Samsung, Microsoft and others seem to have plans in place to ensure Apple's success doesn't happen at their cost.

"If the Vision Pro proves successful and user-friendly, as Apple demonstrated during WWDC 2023, it could drive mass adoption of XR technology like nothing else."

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Chipmakers like Qualcomm have already announced their next-gen spatial computing Snapdragon XR2+ Gen 2, which aims to make blurry and clunky XR interactions a thing of the past. With claimed features including 4.3K per eye resolution, enhanced tracking and environment mapping to unlock more natural and intuitive interactions within XR experiences, optimised performance and battery life while maintaining visual quality allowing for smoother immersive gameplay and longer usage times, a new class of XR devices are coming that will offer seamless overlays of digital information on your surroundings, powered by low-latency video-see-through. See virtual objects interact with your physical surroundings, and feel your real-world gestures mirrored in the digital realm. The possibilities as you can imagine are endless.

Apart from Qualcomm's push, Microsoft's Mesh platform aims to aggressively combine AR and VR experiences with productivity tools, and potentially transform how we work and collaborate. Imagine holding holographic meetings in virtual boardrooms or seamlessly accessing digital information overlaid on real-world environments. In 2024, expect major advancements in Mesh's capabilities, making it a go-to tool for businesses and remote teams.

From an industry standpoint, the ongoing standardisation efforts of the OpenXR Consortium could finally bear fruit in 2024 as well. A unified API and development platform would significantly reduce fragmentation, making it easier for developers to create content for multiple XR devices and platforms. This would attract more talent and investment, leading to a richer and more diverse XR ecosystem. I truly think whatever we saw till 2023 was nothing but proof-of-concepts and niche use cases trying to showcase a few companies' take on what our digital world might look like in AR and VR. But this year, we will really find out how XR experiences will unlock the future of our digital possibilities.